

Yash Pandey

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Vaishali, Ghaziabad, UP 201010

SUMMARY

Results-driven Senior Data Engineer and Data Manager with expertise in architecting enterprise-scale Delta Lakehouses, optimizing cloud-based data pipelines (Azure, AWS), and deploying scalable data governance frameworks. Proven track record of accelerating analytics delivery, centralizing massive datasets, and leading cross-functional engineering teams to build robust, highly available data platforms for financial and enterprise sectors.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, Java, C#, Bash, Node.js
- **Cloud & Data Platforms:** Azure (Synapse Analytics, Data Factory, Data Lake Storage, Logic Apps), AWS (Lambda, QuickSight)
- **Data Architecture & Engineering:** Delta Lakehouse, Medallion Architecture, PySpark, Pandas, Data Pipelines (ETL/ELT), Data Governance, Metadata Management, Data Lineage
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-Learn, BERT, LSTM, NLP, PCA, Multivariate Regression, InferSent, Adversarial Defense (DNE, SEM)
- **DevOps, APIs & Tools:** Git, GitHub Actions, Atlassian Suite (Jira, BitBucket, Confluence), CI/CD, Linux, Flask, REST APIs, Agile

EXPERIENCE

Senior Data Engineer | KlusterFirst | Noida, Uttar Pradesh, India

(April 2026 – Present)

- **Architected** an enterprise-scale Delta Lakehouse to ensure high availability and seamless data flow across 3,000+ daily pipelines for complex client integrations
- **Directed** cross-functional engineering teams as the primary technical contact, translating enterprise business requirements into scalable platforms that accelerated development cycles
- **Designed** robust data governance and metadata management frameworks, ensuring strict compliance and reliable data lineage from Silver to Gold data layers
- **Engineered** automated continuous monitoring and maintenance protocols, drastically reducing administrative overhead and stabilizing data streaming to factory applications

Deputy General Manager - IT | Gokaldas Exports | Bengaluru, Karnataka, India

(March 2025 – April 2026)

- **Centralized** 100M+ records from 30+ disparate sources by designing a metadata-driven Medallion Delta Lakehouse utilizing Azure Data Lake Storage and Synapse Analytics
- **Accelerated** analytics delivery by 60% by engineering scalable Python and PySpark pipelines that eliminated manual reporting bottlenecks
- **Reduced** cross-platform integration errors by 30% by deploying dynamic Python Flask APIs utilizing advanced metaprogramming
- **Established** a centralized single source of truth with automated validation, enabling efficient data reconciliation and federating data responsibility across enterprise teams

Quantitative Programming Specialist | Forefront Analytics | Conshohocken, PA, USA

(September 2022 – August 2024)

- **Developed** automated data solutions and deployed ML models (PCA, multivariate optimization) to drive financial portfolio management strategies
- **Achieved** a 75% reduction in data update latencies (60 to 15 minutes) by redesigning legacy SQL schemas and refactoring extensive codebases for high-frequency market data
- **Boosted** system reliability by 35% by architecting serverless AWS Lambda pipelines and REST APIs for automated database updates
- **Managed** rigorous access controls and security for a Linux-based cryptocurrency validator node, reliably overseeing a \$1M+ stake in a regulated digital asset environment
- **Improved** developer onboarding velocity by 40% by championing Atlassian CI/CD workflows and modernizing internal scripting across three engineering teams

Game Development & QA Co-op | GLIDE Labs | Philadelphia, PA, USA

(September 2020 – March 2021)

- Enhanced player immersion for an AR/VR phone game by designing and developing **3+ hours of interactive gameplay content** partnered with the **Philadelphia Museum of Art**
- Improved the image-scanning feature dataset by automating the **scraping of cycling exhibit images leveraging Python and Selenium WebDriver**
- Resolved **critical login functionality bugs** by authoring and executing **rigorous QA test plans** on the existing Unity3D codebase
- Boosted overall player engagement by ideating and integrating a **thematic user interface overhaul** that aligned with the explorer narrative

Data Scientist Co-op | Glenmede | Philadelphia, PA, USA

(September 2019 – March 2020)

- Accelerated data-driven trade decisions by designing and publishing **interactive AWS QuickSight dashboards** that aggregated and **visualized 45M+ data points**
- Achieved an **87.5% reduction in data extraction runtime** (24 hours to 3 hours) for **100,000+ CUSIP data points** by deploying parallelized, self-modifying Python automation scripts
- Automated investment data collection by developing **web scrapers and scalable AWS Lambda routines**, streamlining the Fixed Income team's workflow, and **reducing extraction workloads**

Software Developer Co-op | IndustrySafe | Philadelphia, PA, USA

(September 2018 – March 2019)

- Improved **SQL database query response times by 40%** by identifying and resolving structural backend bugs within an Agile environment
- Reduced **weekly error report compilation time from 1 hour to seconds** by engineering and executing **automated C# scripts**
- Streamlined OSHA web reporting workflows by developing **dynamic HTML web application forms** tailored to hierarchical employee-manager structures

PROJECTS

Programming Team Lead | Senior Design Project: Rushdown | Philadelphia, PA

(September 2021 – May 2022)

- **Directed** a cross-functional Agile development team to engineer a 3v3 online multiplayer action platformer in Unity 3D (C++), serving as the primary technical liaison between programming, art, and design teams
- **Engineered** core gameplay architectures, integrating the Photon Unity Network for seamless online multiplayer and programming advanced mobility mechanics such as wall-running and sliding
- **Designed** complex player animation controllers utilizing intricate blend trees and motion-captured data to support unique thematic abilities across three distinct characters
- **Optimized** the continuous integration and deployment (CI/CD) pipeline via Perforce and conducted bi-weekly playtest sessions, leveraging statistical feedback graphs to drive iterative feature improvements and periodic builds
- **Streamlined** interdisciplinary collaboration by authoring comprehensive Game Design Documents (GDD), Software Requirements, UMLs, and architecture diagrams to proactively navigate roadblocks and accelerate the development cycle

Programming Team Lead | Momentum | Philadelphia, PA

(April 2021 – September 2021)

- **Engineered** a custom procedural map generation algorithm for a 2D hex-grid strategy game in Unity (C#), building dynamic levels that integrated complex obstacles, puzzles, and strategic enemy placements
- **Designed and implemented** robust enemy AI architectures, programming distinct ranged and melee attack patterns, projectile abilities, and reactive movement systems
- **Streamlined** development workflows by resolving critical source-control and pipeline integration bottlenecks via GitHub, executing rigorous QA testing to ensure seamless art asset deployment
- **Directed** the game's creative narrative and interactive dialogue systems, while concurrently developing a dedicated project website and maintaining comprehensive technical documentation to track weekly progress

Machine Learning Researcher | Textual Adversarial Attacks in NLP | Academic Project | Philadelphia, PA

(March 2022)

- **Evaluated** the robustness of LSTM neural networks against textual adversarial attacks by generating semantic perturbations using the NLPAug Python library
- **Simulated** attacks on sentiment analysis models using clothing review datasets, comparing up-sampling techniques against text augmentation for dataset balancing
- **Improved** model resilience against perturbations by training on augmented data, which increased accuracy under attack from 14.64% to 24.9% and nearly doubled the number of failed attacks (182 to 361)
- **Researched** advanced defense methods, including Dirichlet Neighborhood Ensemble (DNE) and Synonym Encoding Method (SEM), to prevent the misclassification of toxic content and hate speech

EDUCATION & AWARDS

B.S. in Computer Science | Drexel University | Philadelphia, PA | Graduated: June 2022

Concentrations: Artificial Intelligence, Game Programming and Development | **GPA:** 3.36

Honors & Scholarships: A.J. Drexel Scholarship Recipient, Dean's List

Academic Leadership: Teaching Assistant for Systems Architecture (CS281) and Web Development (CS375)

Hackathons & Competitions: Philadelphia Codefest Sponsor Category Winner (2019), 1st Place in Indeed.com "Most Helpful Job Seeker," 1st Place in Comcast Learning Module for Internet Essentials